Discrete Structures

DATE: __/__/__ **Quiz 1 (Section H)**

Q1: $(p\wedge(\sim(\sim p\vee q)))\vee(p\wedge q)\equiv p$

Proof:

LHS:

$\equiv (p\wedge(p\wedge\sim q))\vee(p\wedge q)$ (By De Morgan law)

$\equiv ((p\wedge p)\wedge\sim q)\vee(p\wedge q)$ (By Associative law)

$\equiv (p\wedge\sim q)\vee(p\wedge q)$ (By Idempotent law)

$\equiv p\wedge(\sim q\vee q)$ (By distributive law)

$\equiv p\wedge t$ (By Negation law)

$\equiv p \equiv RHS$ (By Identity law)

Hence proved.

Q2: Write statements in symbolic form.

(a) DATAENDFLAG is off, ERROR equal 0 and SUM is less than 1000

$p\wedge q\wedge r$

(b) DATAENDFLAG is off but ERROR is not equal to 0.

$p\wedge\sim q$

(c) DATAENDFLAG is off; however, ERROR is not 0 or SUM is greater than or equal to 1000

$p\wedge(\sim q\vee\sim r)$

(d) DATAENDFLAG is on and ERROR equals 0 but SUM is greater than or equal to 1000.

$\sim p\wedge q\wedge\sim r$

(e) Either DATAENDFLAG is on or it is the case that both ERROR equals 0 and SUM is less than 1000.

$\sim p\vee(q\wedge r)$

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Q3: Use De Morgan's law to write negations for the statements.

(a): The connector is loose or the machine is unplugged.

Let

p = The connector is loose

q = The machine is unplugged.

$$\boxed{p \vee q}$$

By De Morgan's laws, the negation is

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

The connector is not loose and the machine is plugged

(b): The train is late or my watch is fast.

Let

p = The train is late

q = My watch is fast

$$\boxed{p \vee q}$$

By De Morgan's laws, the negation is:

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

The train is not late and my watch is not fast.